

Munish Patwa

Data Scientist | ML Engineer | Data Analyst

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[LinkedIn](#) | [GitHub](#) | Work Permit: Germany

SUMMARY

M.Sc. Data Science student at TU Dortmund. Built **RAG** pipelines, multi-agent **LLM** systems, end-to-end **ML** forecasting models, and computer vision applications using **Python**, **LangGraph**, **PyTorch**, and **Scikit-Learn**. Experienced with **Docker**, **CI/CD**, **REST APIs**, and model deployment — seeking **Werkstudent** roles in Data Science, ML Engineering, or Data Analytics (available up to 20h/week).

EDUCATION

M.Sc. Data Science | TU Dortmund

Oct 2025 – Present

Dortmund, Germany

Machine Learning, Statistics, Data Mining, Predictive Modelling, Deep Learning

B.Tech. Computer Science – Big Data Analytics | Ganpat University

May 2021 – May 2025

Mehsana, India

GPA: 9.13/10 | ML, Advanced Big Data Analytics, AI, DBMS, Cognitive Computing

WORK EXPERIENCE

AI/ML Engineer Intern | Ascion Global Partners

Jan 2025 – May 2025

Ahmedabad, India (Remote)

- Engineered a production **NLP** chatbot in **Python** handling **3** intent categories (bookings, queries, service requests), reducing average guest query resolution time by **~60%** based on response log benchmarks
- Improved intent classification **F1-score** by **~18%** by training and evaluating **Scikit-Learn** models across **5+** intent classes using a custom preprocessed conversational dataset
- Accelerated model iteration cycles by building a **Matplotlib/Plotly** **EDA** dashboard tracking chatbot usage patterns, error rates, and prediction confidence across test interactions
- Designed an end-to-end ML pipeline from raw data ingestion through **feature** engineering, model training, and evaluation — delivering results in structured reports to cross-functional stakeholders

Tech: Python, NLP, Scikit-Learn, Pandas, Matplotlib, Plotly, Intent Classification, ETL

Teaching Volunteer | Karma Foundation

Jul 2022 – Aug 2022

Ahmedabad, India

- Delivered structured daily Math and English sessions to **15+** students over a **2-week** program; tracked progress via **Microsoft Excel** reports submitted to program coordinators

PROJECTS

Multi-Agent Research System — LangGraph, RAG & LLMs

<https://multiagentresearchwithrag-munishpatwa.streamlit.app/> |

<https://github.com/Munish0303/Multi-Agent-Research-With-RAG>

- Built and deployed a live **5-agent** autonomous research pipeline (**Orchestrator, Researcher, Analyst, Writer, Critic**) on **Streamlit Cloud** using **LangGraph** state machines with a **Critic**-driven critique-and-revise quality loop
- Engineered a dual-mode provider abstraction running fully offline on **Ollama** or in the cloud on **Groq**, auto-selecting the backend at runtime — with a **ChromaDB** **RAG** pipeline supporting PDF/TXT/MD ingestion and per-agent long-term memory
- Integrated multi-provider web search (**Tavily/Brave/DuckDuckGo** with backoff retry), **SQLite** run history, and PDF/Markdown report export; exposed the full pipeline as a **FastAPI REST API** with **async job polling**

Tech: Python, LangGraph, LLM, RAG, ChromaDB, Ollama, Groq, FastAPI, Streamlit, Multi-Agent, Generative AI

YouTube Trending Dashboard — Python, Pandas & Power BI

<https://powerbiyoutubedashboard-munish.streamlit.app/> | <https://github.com/Munish0303/POWERBI-Youtube-Dashboard>

- Built an end-to-end analytics pipeline cleaning **342,656** trending records across **10** countries (from **375,942** raw rows) in **Python (Pandas, ftfy)**, repairing Unicode mojibake and enforcing **0-null** QA assertions before export

- Engineered derived **features** (engagement rate, like/dislike ratio, days-to-trend, views buckets) and designed a 5-page interactive **Power BI** dashboard covering executive KPIs, country comparison, category analysis, channels, and timing/virality
- Deployed a live **Streamlit** gallery sharing the dashboard pages; surfaced insights such as median ~1-day time-to-trend and reach-vs-engagement divergence across content categories

Tech: Python, Pandas, ftfy, Power BI, Streamlit, EDA, Data Visualization, ELT, Data Cleaning

F1 Pit Stop Timing Classifier — XGBoost, SMOTE & SHAP

<https://f1-pitstop-prediction-munish.streamlit.app/> | <https://github.com/Munish0303/F1-Pitstop-Prediction>

- Predicted whether an F1 driver pits on the next lap across **96,336** laps from the **2022-2024** seasons (OpenF1 API), achieving **ROC-AUC 0.979**, **PR-AUC 0.787** (vs. **0.032** baseline), and **F1 0.702** at a tuned threshold — beating Logistic Regression and Random Forest baselines
- Engineered **13** leakage-free **features** (tyre cliff proximity, degradation trend, circuit aggregates fitted on train only) and used a temporal **GroupShuffleSplit** on season; applied **SMOTE** on the **3.1%** minority class after the split to prevent cross-split contamination
- Improved methodological rigour by dropping a dominant but leaky **feature** (in-lap slow-lap flag), cutting its **52%** importance share with **<0.3% ROC-AUC** loss; explained predictions with **SHAP** TreeExplainer and shipped a modular **6-step** pipeline with batch inference

Tech: Python, XGBoost, SMOTE, SHAP, Scikit-Learn, Pandas, Forecasting, Feature Engineering, ETL

FireWatch — Real-Time Fire Detection & Alert System (Docker + CI/CD)

<https://github.com/Munish0303/FireWatch>

- Achieved real-time fire and smoke detection at **<200ms** inference latency by training **YOLOv8** on a custom **1,200-**image **Roboflow** dataset combined with **IoT** sensor fusion
- Containerized the full detection pipeline with **Docker**, deployed via **CI/CD (GitHub Actions)** on **Vercel** with **AWS S3** artifact storage, and automated multi-channel alerts via **SMTP/Twilio**

Tech: YOLOv8, Deep Learning, Computer Vision, Docker, CI/CD, AWS S3, IoT, Next.js, TypeScript, Vercel, REST APIs

SQL Data Cleaning & Analysis — Global Tech Layoffs

https://github.com/Munish0303/MySQL_Mini_Project

- Reduced a **2,361-record** dataset from **~15%** duplicate/null rate to **0%** by building a multi-step **MySQL ELT** pipeline using **CTEs**, window functions, and self-joins for deduplication and backfill
- Identified layoff trend patterns by company, industry, country, and year using rolling totals and dense ranking — delivered findings in an **EDA** report with data quality recommendations

Tech: MySQL, CTEs, Window Functions, EDA, ELT, Data Cleaning, Data Pipelines

SKILLS

Programming: Python, SQL, R, TypeScript, JavaScript

ML / AI: Scikit-Learn, XGBoost, PyTorch, TensorFlow, Keras, Deep Learning, NLP, Computer Vision, SHAP, Forecasting, Feature Engineering, Predictive Modelling, Statistics

LLM & GenAI: LangGraph, LLM Orchestration, RAG, Multi-Agent Systems, Ollama, Groq, ChromaDB, Hugging Face, Prompt Engineering, Generative AI

Data & Analytics: Pandas, NumPy, Spark/PySpark, Power BI, Tableau (familiar), Matplotlib, Seaborn, Plotly, EDA, Data Cleaning, SMOTE, ETL, ELT, Data Pipelines, Probability Theory

MLOps & Cloud: Docker, CI/CD (GitHub Actions), MLOps, Kubernetes (familiar), AWS S3/EC2 (familiar), GCP Vertex AI (familiar), REST APIs, FastAPI, Vercel, Firebase, Git, GitHub

Databases: MySQL, PostgreSQL, Supabase

Web & Tools: React, Next.js, Streamlit, Microsoft Excel, Roboflow, Kaggle

Languages: English (C1), German (A2 — improving), Gujarati (Native), Hindi (C1)